Reviewer Pack — Minimal Rank of Tropical Hermitian Forms vs Communication Co…

Entry c00e2462e5e0 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

2026-05-25 17:29:52 UTC

Minimal Rank of Tropical Hermitian Forms vs Communication Complexity for DISJOINTNESS

Entry ID: `c00e2462e5e0` **Verdict**: `INCONCLUSIVE` **Recorded**: 2026-05-25 17:29:52 UTC

1. Conjecture

Field A (mathematical branch): Tropical Geometry × Tropical Algebra **Field B** (complexity object): Communication Complexity: DISJOINTNESS

Statement:

> For any n-ary Boolean function f, the minimal rank τ(f) of its corresponding tropical Hermitian form is lower bounded by Ω(n^(1/2)) for all instances with communication complexity greater than n^(1/4). Equivalently, for all functions f with C(f) > n^(1/4), τ(f) = Ω(n^(1/2)).

Rationale (proposer's reasoning):

> Tropical Hermitian forms provide a geometric interpretation of Boolean functions that could potentially reveal hidden structures in their complexity. The conjecture suggests that such an invariant captures the essence of communication complexity for disjointness, which is known to be hard.

Taxonomy category: `COMM_DISJ` (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): `00939e96494e4c41`

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

> If the mean minimal rank τ(f) of tropical Hermitian forms for random n-ary Boolean functions f with communication complexity C(f) > n^(1/4) is greater than or equal to Ω(n^(1/2)), then the conjecture is supported. If any seed produces a case where τ(f) < Ω(n^(1/2)) for a function f with C(f) > n^(1/4), the conjecture is falsified.

3. Barrier filter (F1)

Final: `PASS`

Barrier	Verdict	Confidence	LLM_1	LLM_2
RELATIVIZATION	UNCERTAIN	1.00	UNCERTAIN	HITS
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	UNCERTAIN	1.00	UNCERTAIN	HITS
KARP_LIPTON	SAFE	0.95	SAFE	UNCERTAIN

4. Novelty audit

Verdict: `NOVEL` against 9 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - `tropical geometry tropical algebra AND communication complexity disjointness` - `minimal rank tropical Hermitian forms AND communication complexity DISJOINTNESS` - `Ω(n^(1/2)) lower bound minimal rank AND communication complexity greater than n^(1/4)`

Top relevant hits considered: - [http://arxiv.org/abs/1207.2443v2] Tropical Teichmuller and Siegel spaces - [http://arxiv.org/abs/1204.3875v2] Tropicalizing vs Compactifying the Torelli morphism - [http://arxiv.org/abs/1206.1925v1] Counting Algebraic Curves with Tropical Geometry - [http://arxiv.org/abs/1610.00298v4] Khovanskii bases, higher rank valuations and tropical geometry - [http://arxiv.org/abs/2510.17487v1] Directional Search for Persistent Gravitational Waves: Results from the First Part of LIGO-Virgo-KAGRA's Fourth Observin - [http://arxiv.org/abs/1411.4413v2] Observation of the rare $B_s^0 \to \mu^+\mu^-$ decay from the combined analysis of CMS and LHCb data - [http://arxiv.org/abs/2601.07595v3] Deep Search for Joint Sources of Gravitational Waves and High-Energy Neutrinos with IceCube During the Third Observing R - [s2:6f234602ceea6022678a355ae683962b95691547] The intersection of commutative algebra and algebraic geometry with combinatorics and discrete geometry is a rich and ex

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)
Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution**: rc=1, elapsed=0.1s

5.1 Generated Python source

```python
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_random_function(n):
        return [random.choice([0, 1]) for _ in range(2**n)]

    def communication_complexity(f):
        n = int(math.log2(len(f)))
        max_comm = 0
        for i in range(2**n):
            for j in range(i+1, 2**n):
                if f[i] != f[j]:
                    comm = bin(i ^ j).count('1')
                    if comm > max_comm:
                        max_comm = comm
        return max_comm

    def tropical_hermitian_form(f):
        n = int(math.log2(len(f)))
```

```
        H = [[0 for _ in range(n)] for _ in range(n)]
        for i in range(2**n):
            for j in range(i+1, 2**n):
                if f[i] != f[j]:
                    comm = bin(i ^ j).count('1')
                    H[comm][i % n] += 1
                    H[comm][j % n] += 1
        return H

    def min_rank(H):
        n = len(H)
        rank = 0
        for i in range(n):
            if any(H[j][i] != 0 for j in range(rank, n)):
                rank += 1
                for j in range(n):
                    if H[j][i]:
                        for k in range(n):
                            H[j][k] -= H[i][k]
        return rank

    n_values = [5, 10, 15, 20, 30, 40]
    total_rank = 0
    instances_tested = 0

    for n in n_values:
        f = generate_random_function(n)
        comm = communication_complexity(f)
        if comm > n**(1/4):
            H = tropical_hermitian_form(f)
            rank = min_rank(H)
            total_rank += rank
            instances_tested += 1

    mean_rank = total_rank / instances_tested if instances_tested > 0 else 0
    conjecture_holds = mean_rank >= math.sqrt(n_values[-1])

    return {
        "metric_name": "min_rank",
        "metric_value": mean_rank,
        "instances_tested": instances_tested,
        "conjecture_holds": conjecture_holds,
        "counterexample": "" if conjecture_holds else f"mean_rank={mean_rank} < sqrt({n_values[-1]
    }

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29] * 3

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_rank = sum(r["metric_value"] for r in results) / len(results)
```

```
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_rank} std=0.0 support_fraction=1.0")
    elif support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_rank} std={math.sqrt(sum((r['metric_value'] - mean_ra
    else:
        first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample=\"mean_rank<{mean_rank} < sqrt({n_values[-1]})\"
```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e3793602.py", line 88, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e3793602.py", line 66, in run_trial
    H = tropical_hermitian_form(f)
        ^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e3793602.py", line 42, in tropical_herm
    H[comm][i % n] += 1
    ~^^^^^^
IndexError: list index out of range
```

8. Critic adversarial review

Critic verdict: `CHALLENGE`

Critic reasoning:

> skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: `INCONCLUSIVE`

Reasoning:

> The test code crashed before producing data, which means it did not complete its execution to verify the conjecture. | next: Re-run the test with proper error handling and ensure that it completes without crashing.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	10033
2	preregistration	ollama_remote	glm4:latest	0	0	6408
3	novelty	ollama_remote	glm4:latest	0	0	4706
4	novelty	ollama_remote	glm4:latest	0	0	6084

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	19624
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8857
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10993
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10281
9	judge	ollama_remote	glm4:latest	0	0	14771

Totals: 0 input tokens, 0 output tokens, ~$0.0000 cost, 91757 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in `research/audit/c00e2462e5e0.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/c00e2462e5e0.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)
Replay tarball: `research/replay/c00e2462e5e0.tar.gz` (if generated)